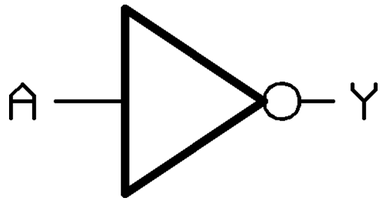
Portas Lógicas Básicas

Circuitos (Digitais) Combinacionais

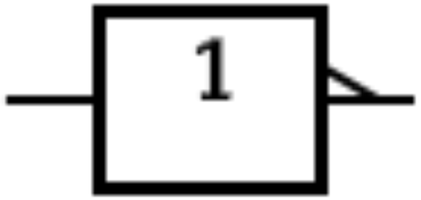
Prof. Fernando Passold (c) 2026

Porta NOT (Inversora)

- Símbolo ANSI/IEEE:



- Símbolo IEC:



- Tabela Verdade:

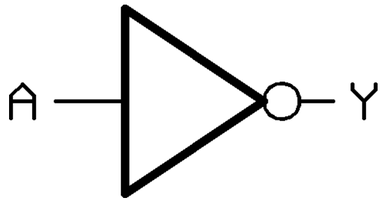
Entrada A	Saída Y
0	1
1	0

- Equação (booleana):

$$Y = \overline{A}$$

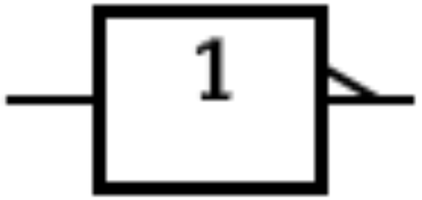
Porta NOT (Inversora)

- Símbolo ANSI/IEEE:



Símbolos de **formato distintivo** (*distinctive-shape*)
IEEE Std 91
ANSI Y32.14
IEEE/AIEE 91-1962
Início: 1953 (diretriz militar MIL-STD_806)
Última atualização: **1984** (Reafirmado em **1994** no IEEE 91)
(Não é um parda internacional, mas tolerado)

- Símbolo IEC:

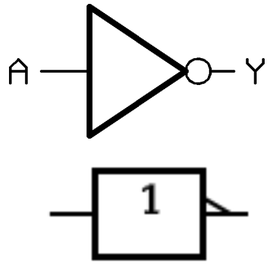


Símbolos de **formato retangular** (*rectangular-shape*):
ANSI/IEEE Std 91-1984
(IEEE Standard Graphics Symbols for Logic Functions, em conjunto com o American National Standards Institute)
Padrão internacional IEC 60617-12
(International Electrotechnical Commission)
Início: 1984 (Como norma conjunta ANSI/IEEE Std 91-1984)
Última atualização: **1991** (Adendo IEEE 91a) / **1997** (IEC 60617-12)

Status Atual: Ambos os padrões estão maduros e não sofrem alterações profundas desde a década de 1990, pois a indústria migrou o design de circuitos para **linguagens de descrição de hardware (como VHDL e Verilog)**, tornando os diagramas de portas lógicas puramente educacionais ou restritos a blocos funcionais de alto nível.

Linguagens VHDL e Verilog...

Mercado de FPGAs, mundo acadêmico, Setor militar, aeroespacial, Telecom, infra. Indusrtrial



Maior adoção ASICs, Processadores

VHDL:

```
-- Importa a biblioteca padrão para tipos de dados lógicos
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Define as entradas e saídas do componente
entity porta_not is
    Port (
        entrada : in  std_logic;
        saida   : out std_logic
    );
end porta_not;

-- Define o comportamento interno do componente
architecture Comportamental of porta_not is
begin
    -- Atribuição concorrente (o sinal da saída é atualizado
se a entrada mudar)
    saida <= not entrada;
end Comportamental;
```

Verilog: Usa o operador de negação bit a bit ~, ou primitiva not:

```
module porta_not (
    input  entrada,
    output saida
);
    // Atribuição contínua usando o operador bitwise NOT (~)
    assign saida = ~entrada;

endmodule
```

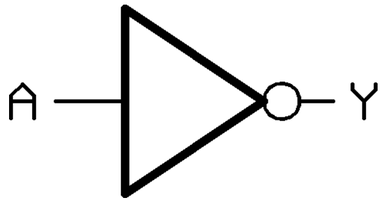
```
module porta_not_estrutural (
    input  entrada,
    output saida
);
    // Primitiva nativa: nome_da_porta (saída, entrada);
    not instancia1 (saida, entrada);

endmodule
```

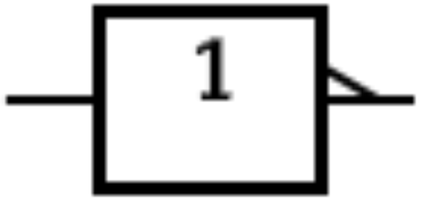
SystemVerilog: versão expandida do Verilog (designs de CIs, ASICs — indústria).

Porta NOT (Inversora)

- Símbolo ANSI/IEEE:



- Símbolo IEC:



- Tabela Verdade:

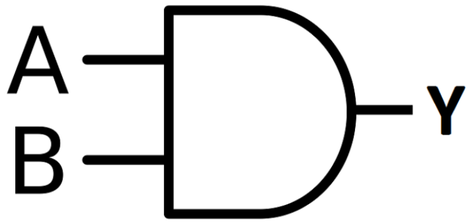
Entrada A	Saída Y
0	1
1	0

- Equação (booleana):

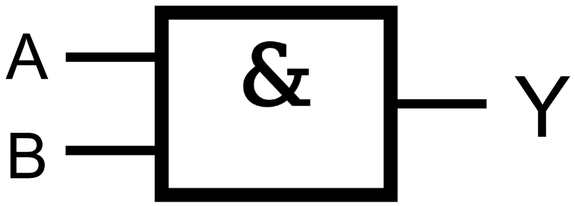
$$Y = \overline{A}$$

Porta AND

- Símbolo ANSI/IEEE:



- Símbolo IEC:



- Tabela verdade

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Note: $x \cdot 0 = 0$

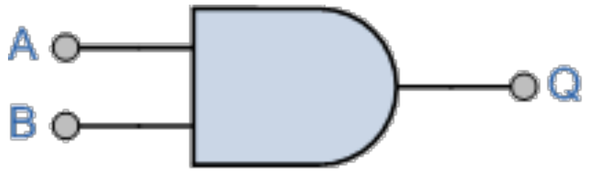
$x \cdot 1 = x$

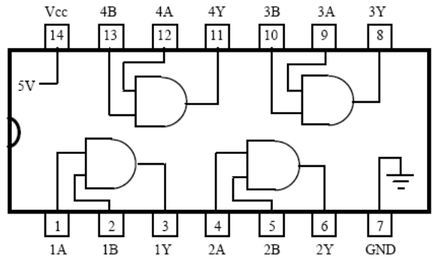
$1 \cdot 1 = 1$

- Equação:

$$Y = A \cdot B$$

- Note:



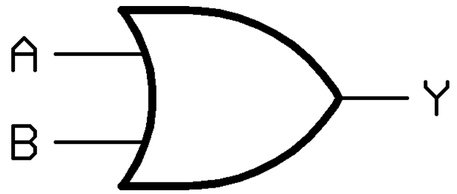


74LS08

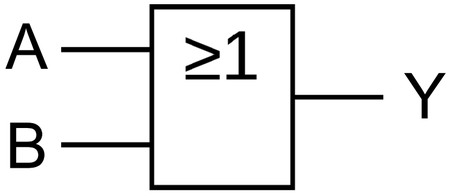
Obs.: entradas > 1 / saídas = 1.

Porta OR

- Símbolo ANSI/IEEE:



- Símbolo IEC:



- Tabela verdade

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Note: $x + 1 = 1$

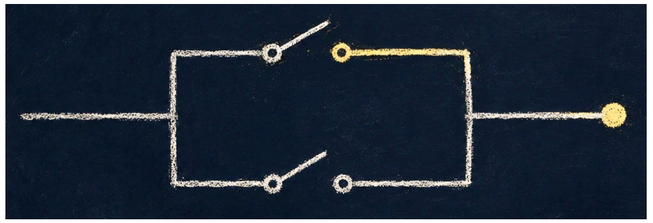
$x + 0 = x$

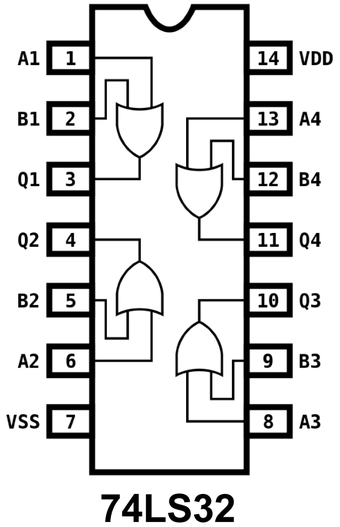
$0 + 0 = 0$

- Equação:

$$Y = A + B$$

- Note:





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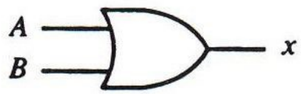
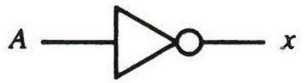
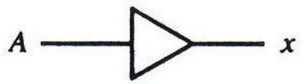
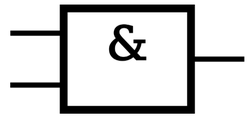
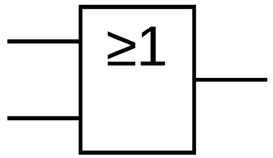
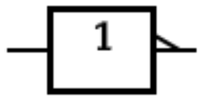
Obs.: entradas > 1 / saídas=1.

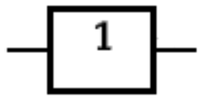
Primeiro Resumo

Name	*Graphic symbol*	*Algebraic function*	*Truth table*
AND		$x = A \cdot B$ or $x = AB$	A B | x 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1
OR		$x = A + B$	A B | x 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 1
Inverter		$x = A'$	A | x 0 | 1 1 | 0
Buffer		$x = A$	A | x 0 | 0 1 | 1



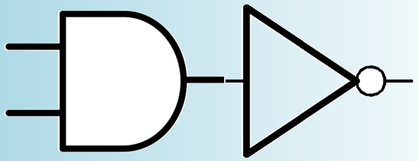




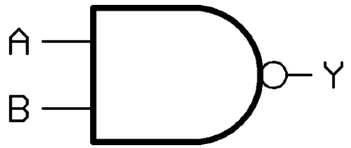


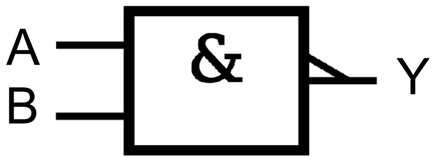
Variações

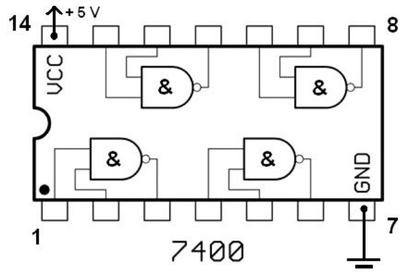
AND + NOT

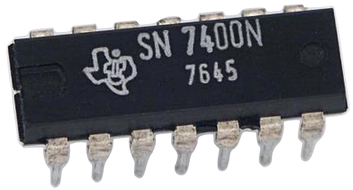


NAND





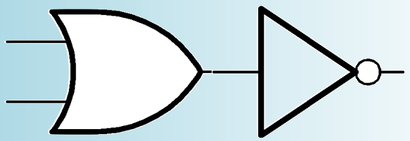




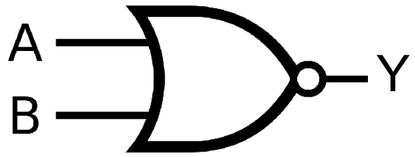
NAND

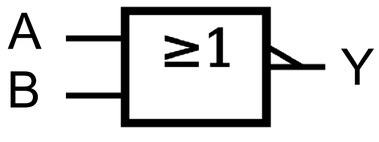
A B	AND $= A \cdot B$	NAND $Y = \overline{A \cdot B}$	Y
0 0	0	1	$Y = \overline{0 \cdot 0} = \overline{0} = 1$
0 1	0	1	$Y = \overline{0 \cdot 1} = \overline{0} = 1$
1 0	0	1	$Y = \overline{1 \cdot 0} = \overline{0} = 1$
1 1	1	0	$Y = \overline{1 \cdot 1} = \overline{1} = 0$

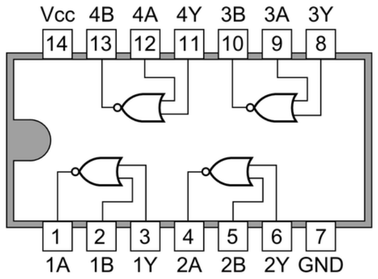
OR + NOT



NOR







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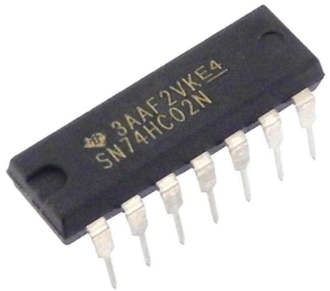


NOR

A B	OR $= A + B$	NOR $Y = \overline{A + B}$	Y
0 0	0	1	$Y = \overline{0 + 0} = \overline{0} = 1$
0 1	1	0	$Y = \overline{0 + 1} = \overline{1} = 0$
1 0	1	0	$Y = \overline{1 + 0} = \overline{1} = 0$
1 1	1	0	$Y = \overline{1 + 1} = \overline{1} = 0$

Resumo

A B	AND	OR	NAND	NOR
0 0	0	0	1	1
0 1	0	1	1	0
1 0	0	1	1	0
1 1	1	1	0	0
	$x = A \cdot B$	$x = A + B$	$x = \overline{A \cdot B}$	$x = \overline{A + B}$

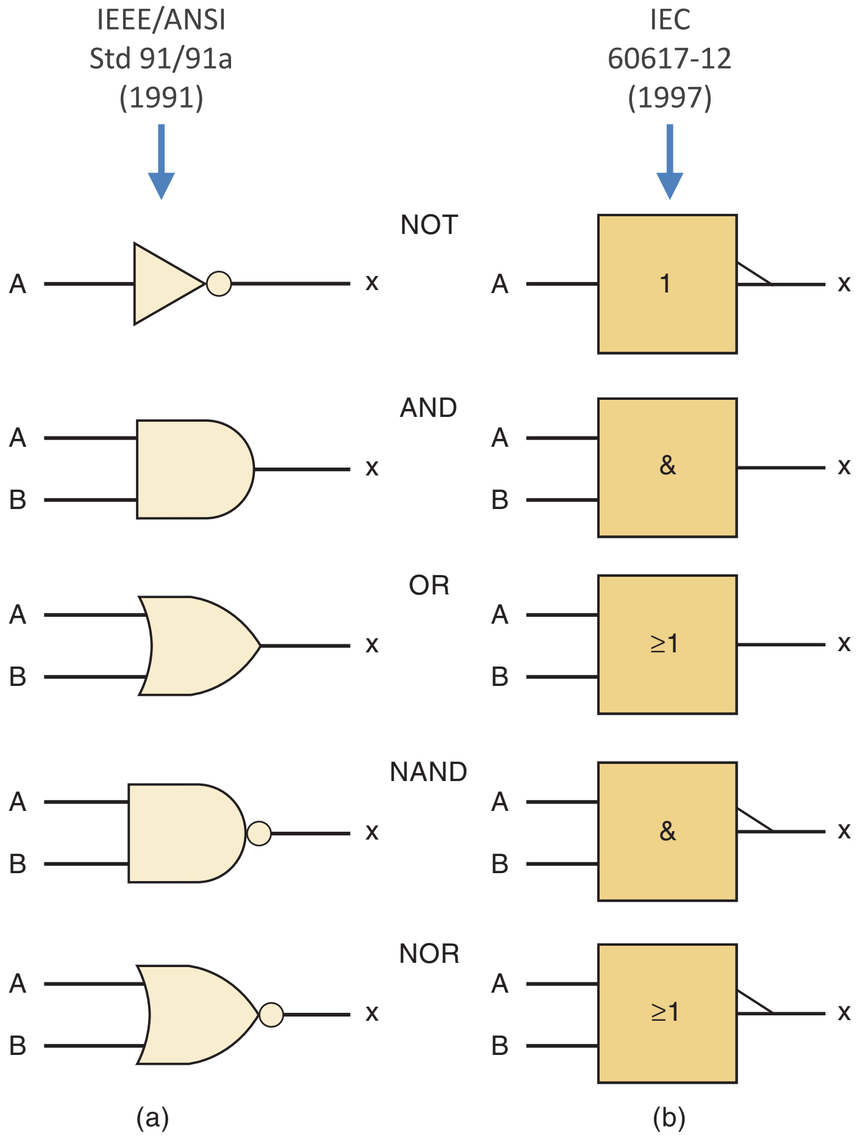
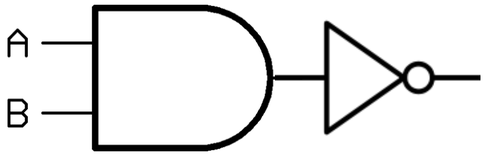


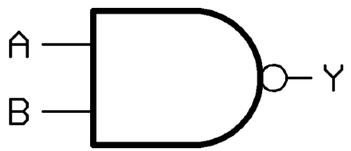
FIGURE 3-41 Standard logic symbols: (a) traditional; (b) IEEE/ANSI.

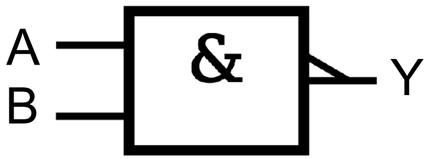
Atenção

AND + NOT



NAND



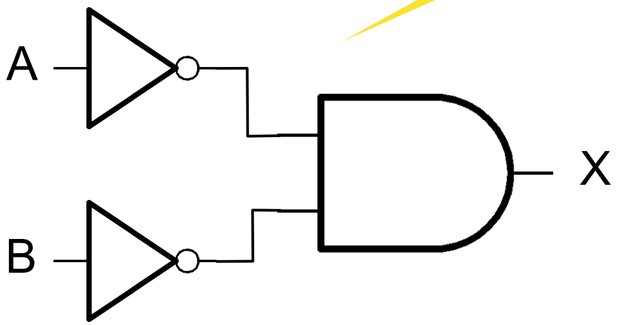


$\therefore$

Equação NAND:

$Y = \overline{A \cdot B}$

Note porém, que: $X = \overline{A} \cdot \overline{B} \neq Y = \overline{A \cdot B}$

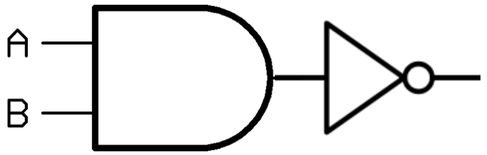


NAND:

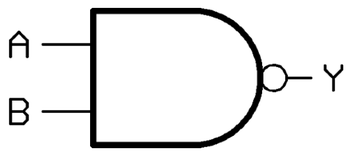
A	B	Y	AND
0	0	$Y = \overline{0 \cdot 0} = \overline{0} = 1$	0
0	1	$Y = \overline{0 \cdot 1} = \overline{0} = 1$	0
1	0	$Y = \overline{1 \cdot 0} = \overline{0} = 1$	0
1	1	$Y = \overline{1 \cdot 1} = \overline{1} = 0$	1

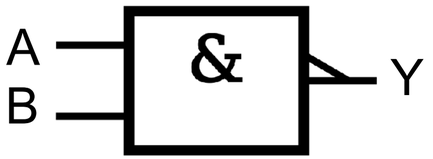
Atenção

AND + NOT



NAND



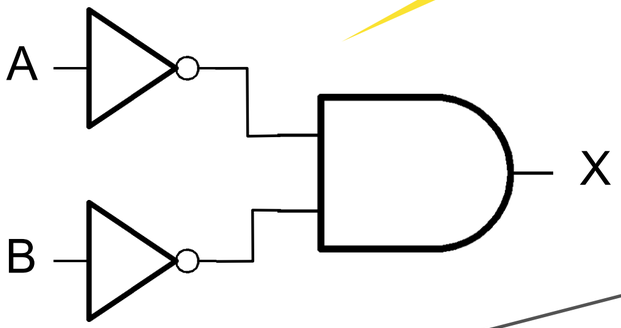


$\therefore$ **Equação NAND:** $Y = \overline{A \cdot B}$

NAND:

A	B	Y	AND
0	0	$Y = \overline{0 \cdot 0} = \overline{0} = 1$	0
0	1	$Y = \overline{0 \cdot 1} = \overline{0} = 1$	0
1	0	$Y = \overline{1 \cdot 0} = \overline{0} = 1$	0
1	1	$Y = \overline{1 \cdot 1} = \overline{1} = 0$	1

Note porém, que: $X = \overline{A} \cdot \overline{B} \neq Y = \overline{A \cdot B}$

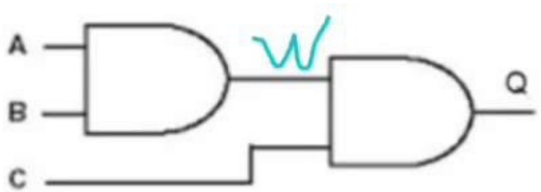


NOR ⚠

A	B	X
0	0	$X = \overline{A} \cdot \overline{B} = \overline{0} \cdot \overline{0} = 1 \cdot 1 = 1$
0	1	$X = \overline{A} \cdot \overline{B} = \overline{0} \cdot \overline{1} = 1 \cdot 0 = 0$
1	0	$X = \overline{A} \cdot \overline{B} = \overline{1} \cdot \overline{0} = 0 \cdot 1 = 0$
1	1	$X = \overline{A} \cdot \overline{B} = \overline{1} \cdot \overline{1} = 0 \cdot 0 = 0$

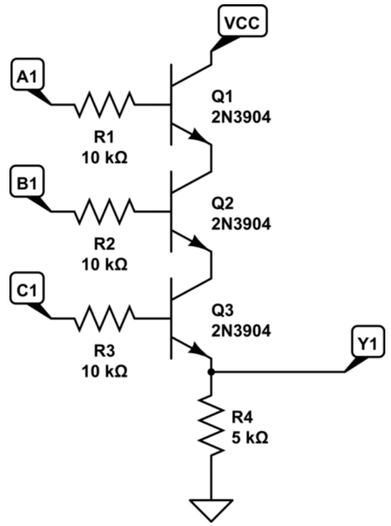
$\neq$

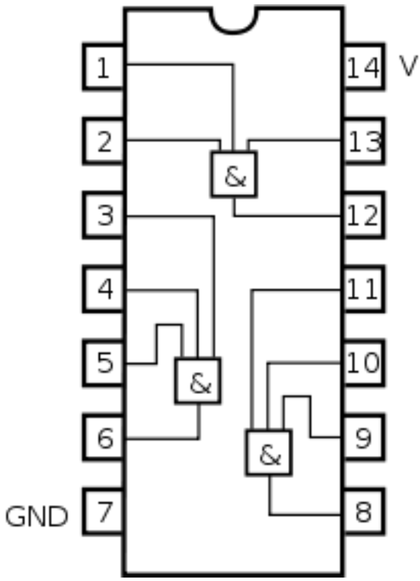
Análise Circuitos: Exemplos:

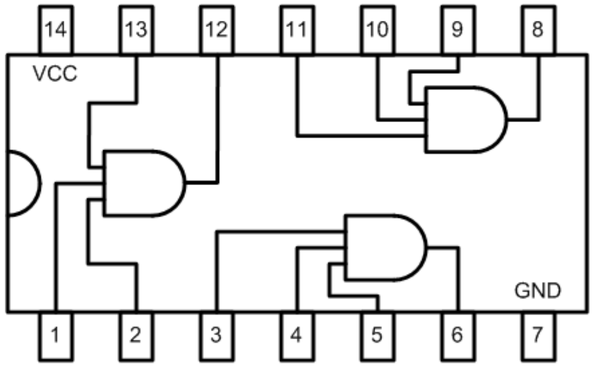


Equação ?

3 input AND				
Input				Output
A	B	C		Q
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

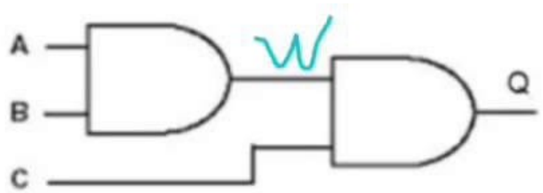






7411 Triple 3 Input AND

Análise Circuitos: Exemplos:



3 input AND				
Input				Output
A	B	C	W	Q
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1

Equação ?

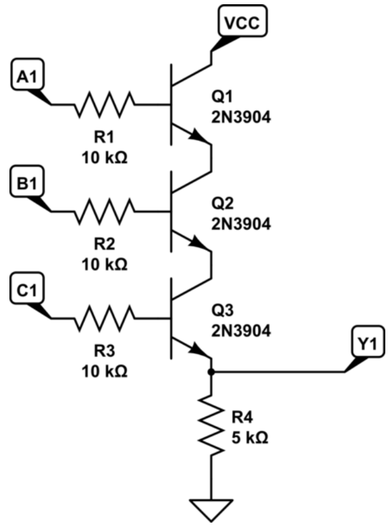
$$Q = A \cdot B \cdot C$$

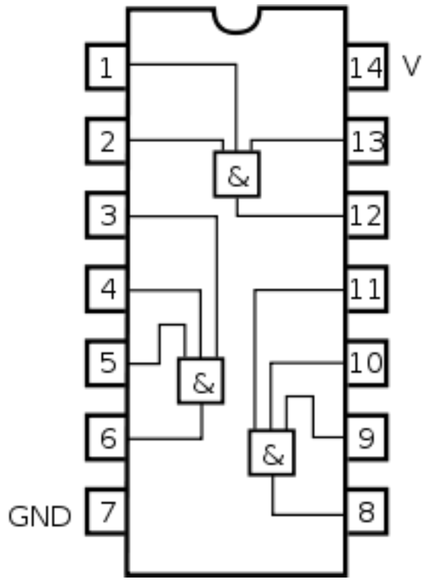
Note:

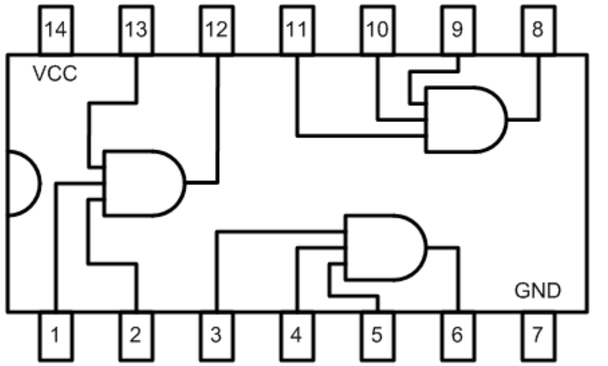
Se $A = 0 \therefore Q = 0 \cdot B \cdot C = 0$

e

$$Q = 1 \iff A = 1, B = 1, C = 1$$
$$\Rightarrow Q = 1 \cdot 1 \cdot 1 = 1$$

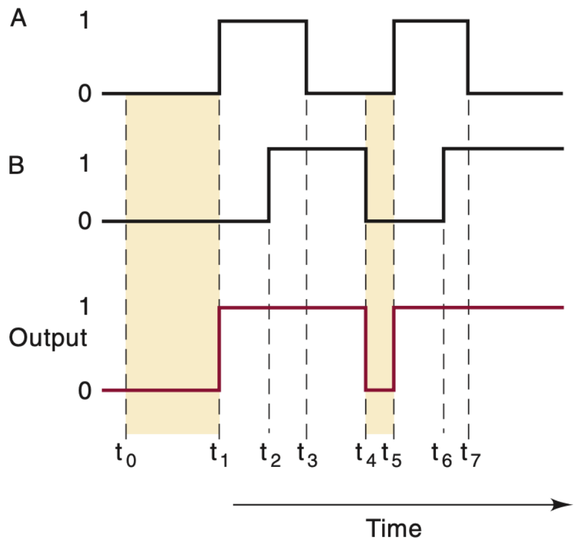




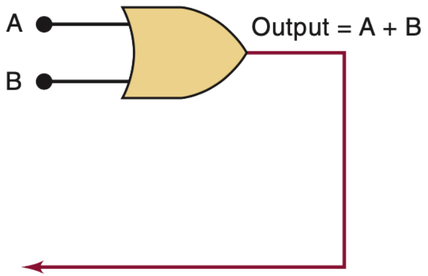


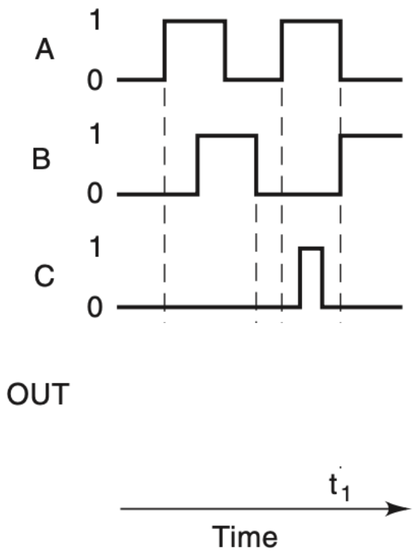
7411 Triple 3 Input AND

Análise Circuitos: Exemplos:

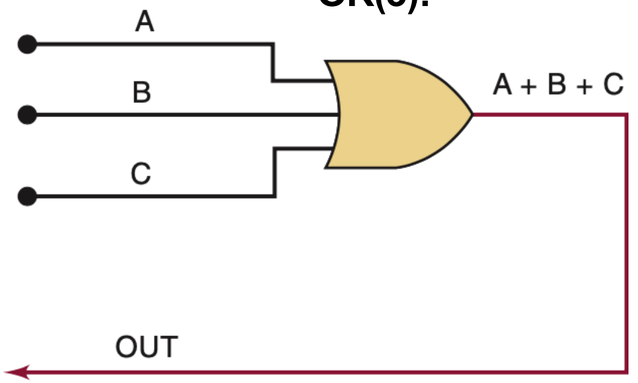


Porta OR(2):

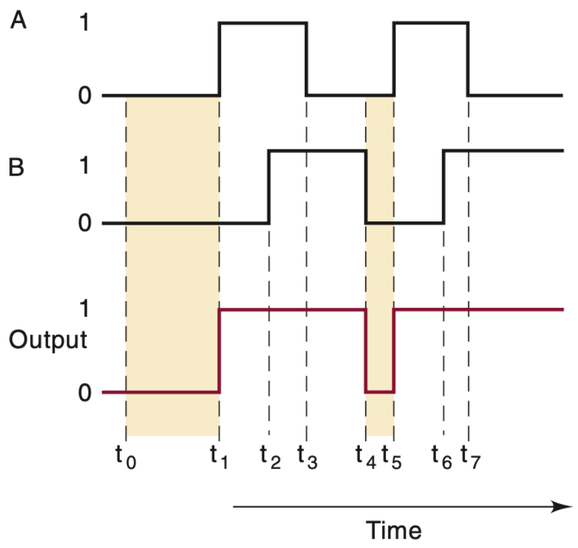




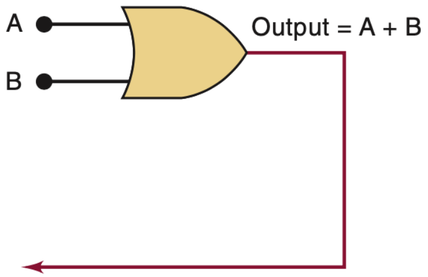
Porta OR(3):

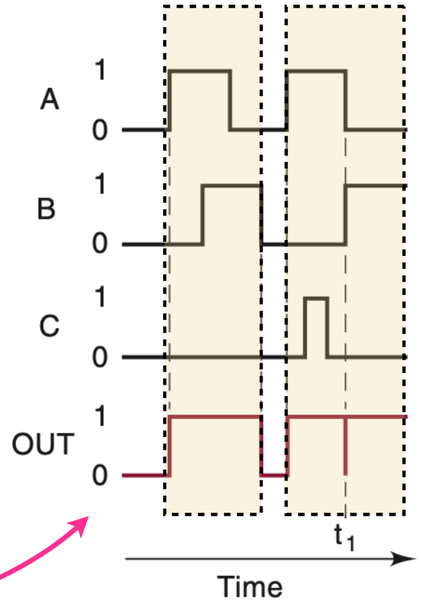


Análise Circuitos: Exemplos:

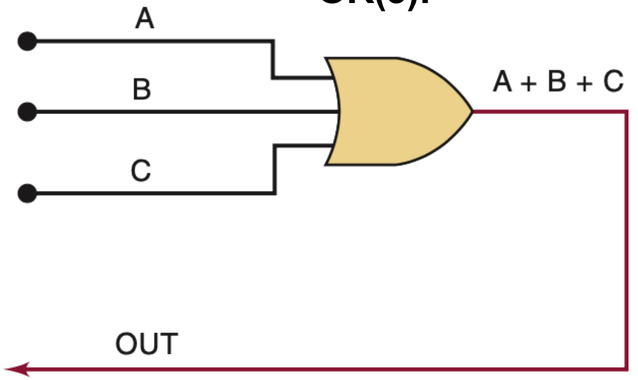


Porta OR(2):





Porta OR(3):



Porta OR:

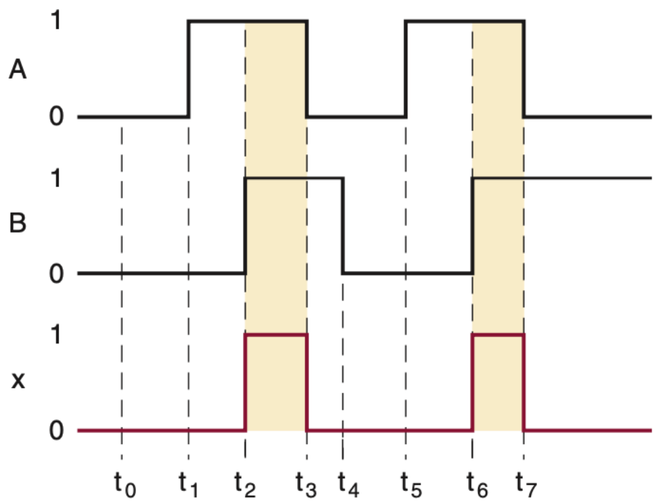
Note: basta uma entrada em nível lógico ALTO para saída comutar para nível lógico ALTO:

Álgebra de Boole:

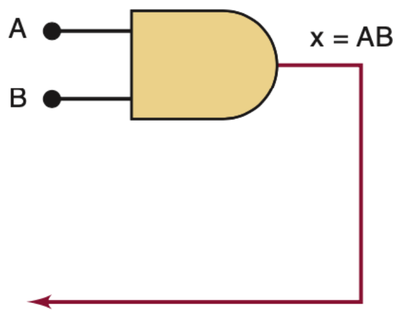
$$x + 0 = x$$

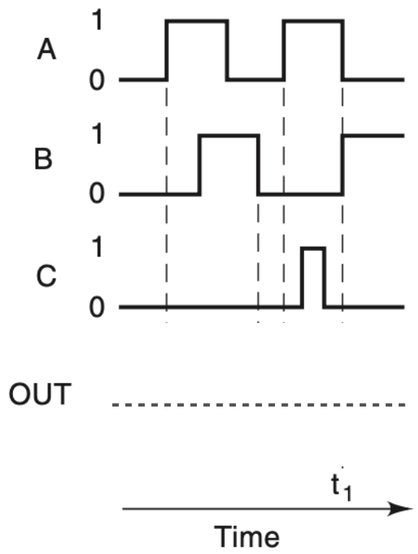
$$x + 1 = 1$$

Análise Circuitos: Exemplos:

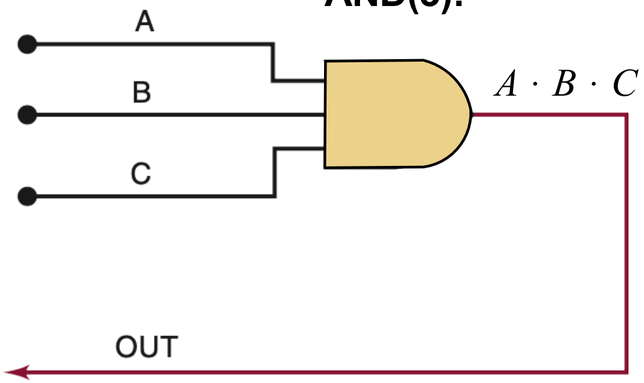


Porta AND(2):

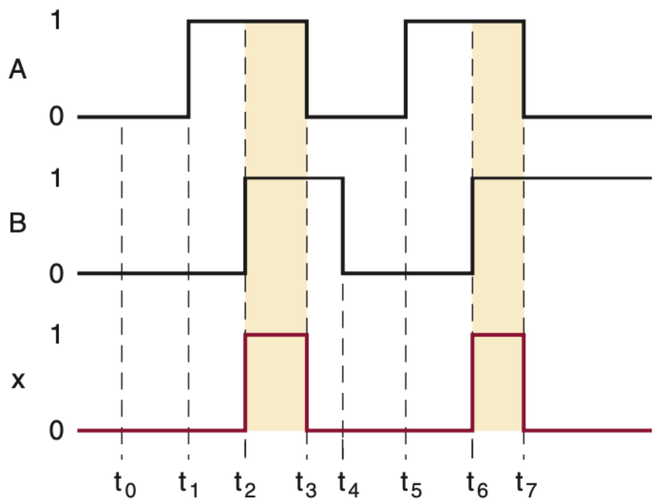




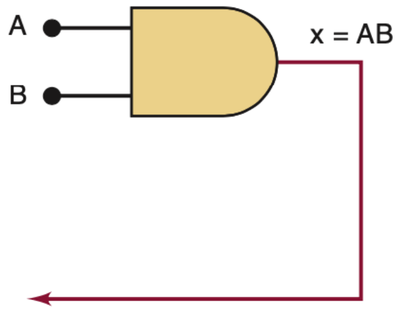
Porta AND(3):

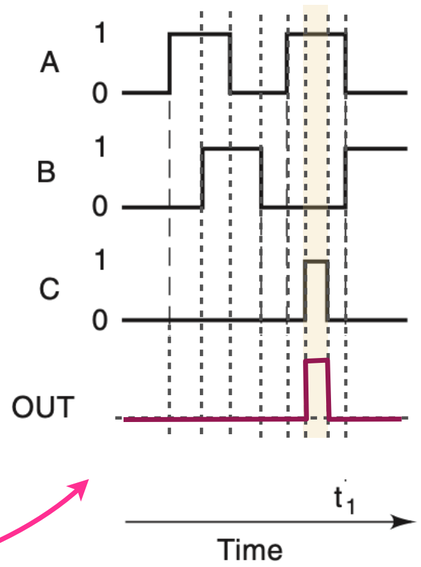


Análise Circuitos: Exemplos:

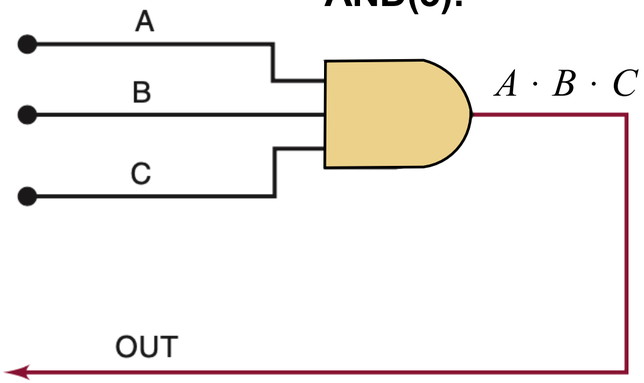


Porta AND(2):





Porta AND(3):



Porta AND:

Note: basta uma entrada em nível lógico BAIXO para saída comutar para nível lógico BAIXO

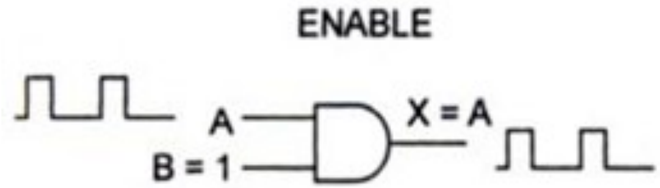
Álgebra de Boole:

$$x \cdot 0 = 0$$

$$x \cdot 1 = x$$

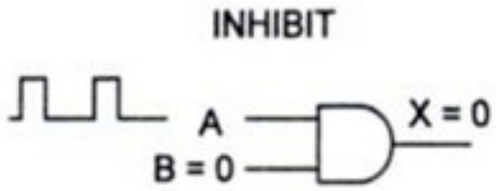
Intro à Álgebra de Boole:

Porta AND:



Note:

$$x \cdot 1 = x$$



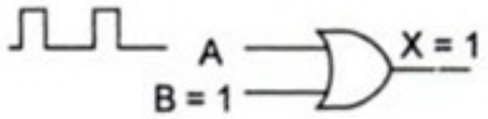
$$x \cdot 0 = 0$$

Obs.: A porta AND pode funcionar como uma “chave" com Entrada de “Enable"

Porta OR:

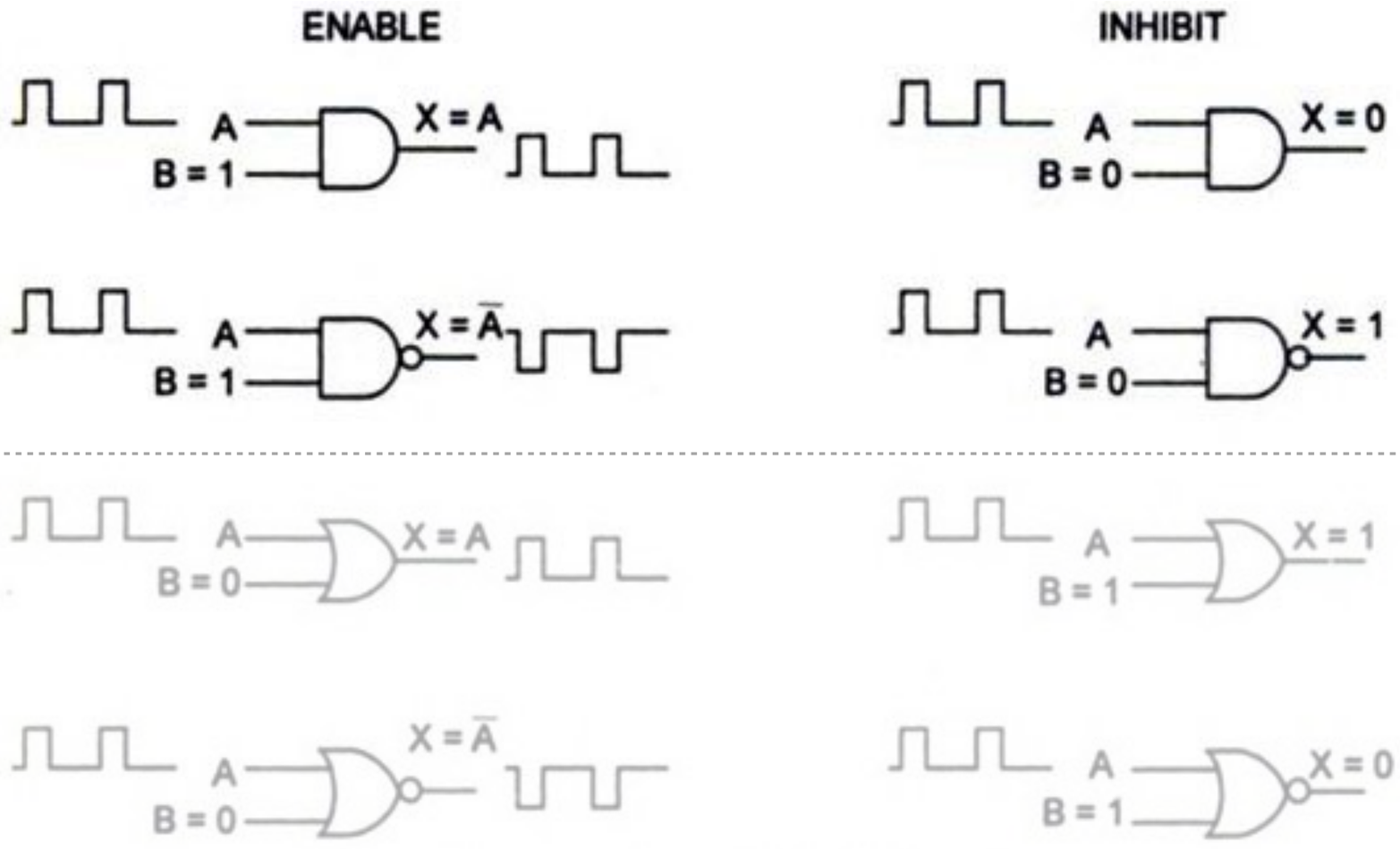
Note:

$$x + 1 = 1$$



$$x + 0 = x$$

Entrada Enable



Porta AND:
Note:

$x \cdot 0 = 0$

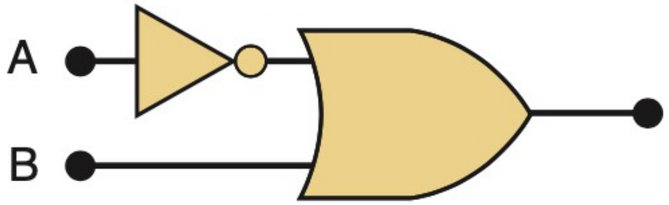
$x \cdot 1 = x$

Porta OR:
Note:

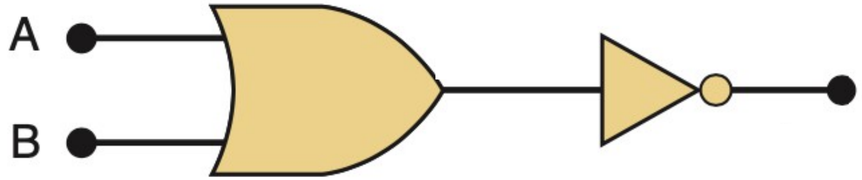
$x + 0 = x$

$x + 1 = 1$

Análise Circuitos: Exemplos:



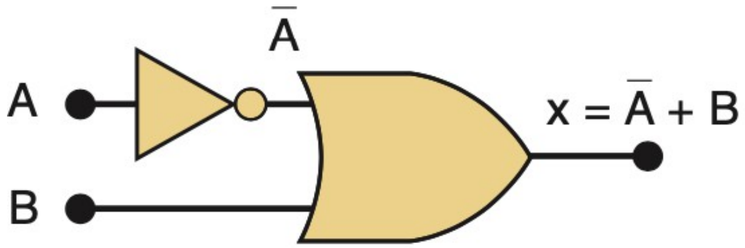
(a)



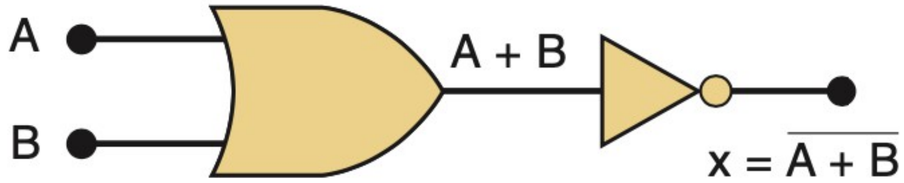
(b)

A B	(a)	A B	(b)
0 0		0 0	
0 1		0 1	
1 0		1 0	
1 1		1 1	

Análise Circuitos: Exemplos:



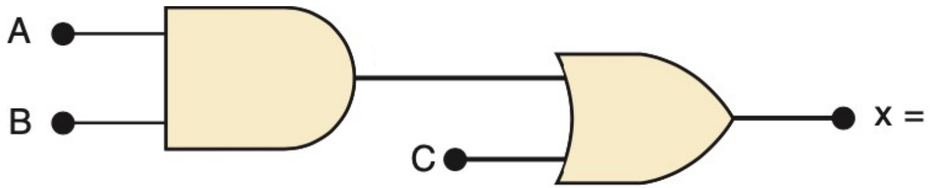
(a)



(b)

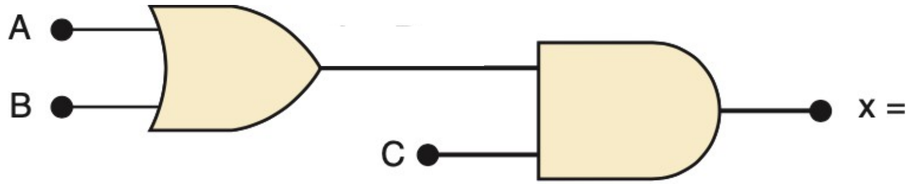
A B	(a)	A B	(b)
0 0		0 0	
0 1		0 1	
1 0		1 0	
1 1		1 1	

Análise Circuitos: Exemplos:



(a)

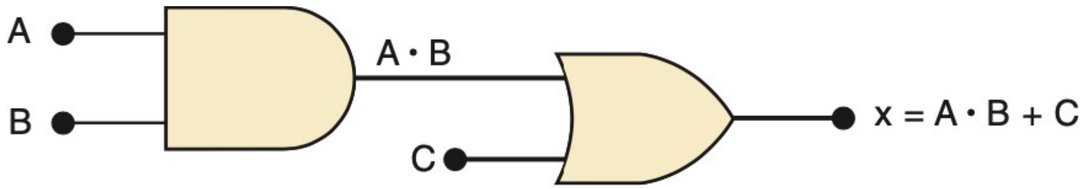
Ref	A	B	C	A · B	x
0					
1					
2					
3					
4					
5					
6					
7					



(b)

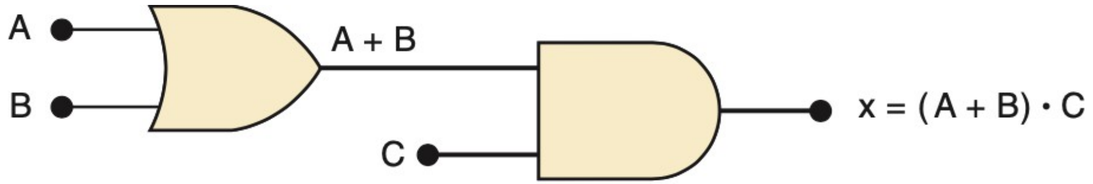
Ref	A	B	C	A + B	x
0					
1					
2					
3					
4					
5					
6					
7					

Análise Circuitos: Exemplos:



(a)

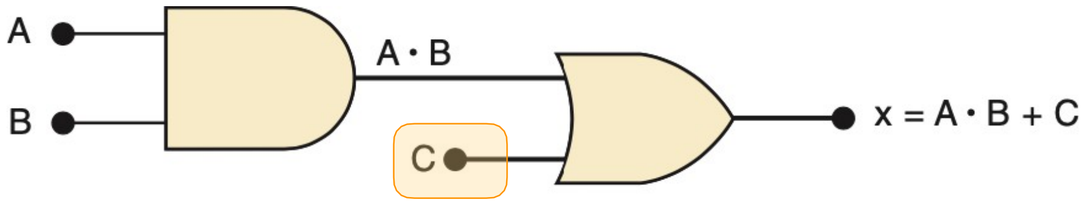
Ref	A	B	C	A · B	x
0	0	0	0		
1	0	0	1		
2	0	1	0		
3	0	1	1		
4	1	0	0		
5	1	0	1		
6	1	1	0		
7	1	1	1		



(b)

Ref	A	B	C	A + B	x
0	0	0	0		
1	0	0	1		
2	0	1	0		
3	0	1	1		
4	1	0	0		
5	1	0	1		
6	1	1	0		
7	1	1	1		

Análise Circuitos: Exemplos:

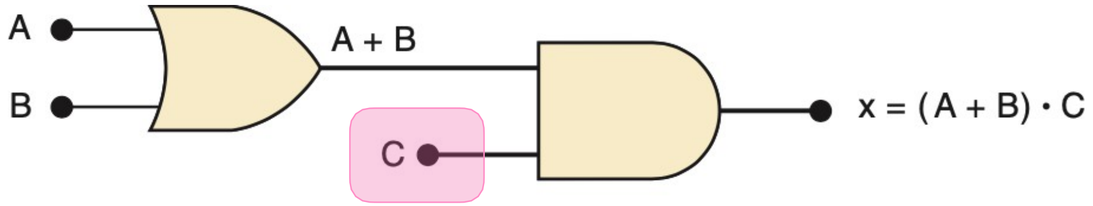


(a)

Ref	A	B	C	A · B	x
0	0	0	0		
1	0	0	1		1
2	0	1	0		
3	0	1	1		1
4	1	0	0		
5	1	0	1		1
6	1	1	0		
7	1	1	1		1

$x + 1 = 1$

Porta OR: Basta uma entrada = “1”, Saída = “1"



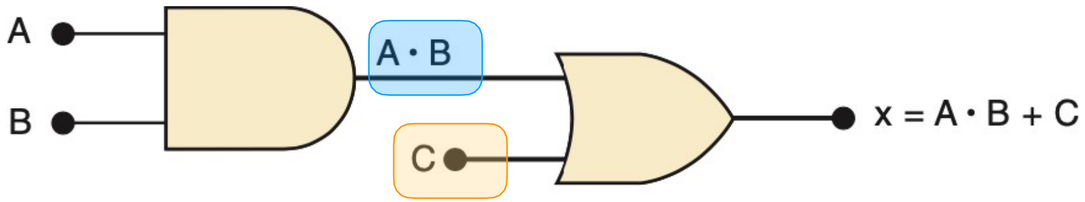
(b)

$x \cdot 0 = 0$

Ref	A	B	C	A + B	x
0	0	0	0		0
1	0	0	1		
2	0	1	0		0
3	0	1	1		
4	1	0	0		0
5	1	0	1		
6	1	1	0		0
7	1	1	1		

Porta AND: Basta uma entrada = “0”, Saída = “0”

Análise Circuitos: Exemplos:



(a)

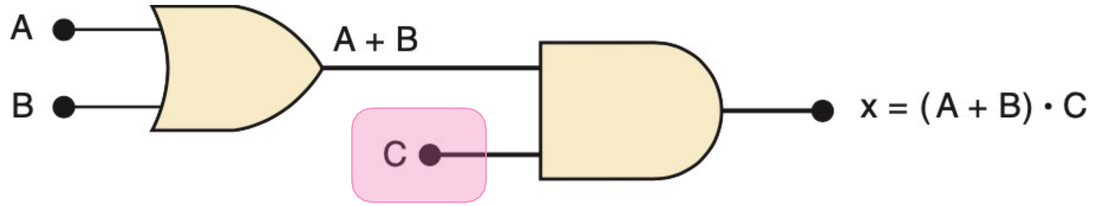
Ref	A	B	C	A · B	x
0	0	0	0	0	0
1	0	0	1	0	1
2	0	1	0	0	0
3	0	1	1	0	1
4	1	0	0	0	0
5	1	0	1	0	1
6	1	1	0	1	1
7	1	1	1	1	1

$x + 1 = 1$

Porta OR:
Basta uma entrada = “1”,
Saída = “1"

Saída da AND = “1" apenas Quando TODAS as entradas = “1".

$1 \cdot 1 \iff 1$



(b)

$x \cdot 0 = 0$

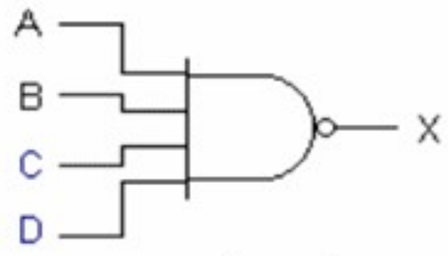
Ref	A	B	C	A + B	x
0	0	0	0		0
1	0	0	1		0
2	0	1	0		0
3	0	1	1		1
4	1	0	0		0
5	1	0	1		1
6	1	1	0		0
7	1	1	1		1

Porta AND:
Basta uma entrada = “0”,
Saída = “0”

Porta OR:
Basta uma entrada = “1”,
Saída = “1"

Exemplo:

- Que porta é esta?



$$X = \overline{ABCD}$$

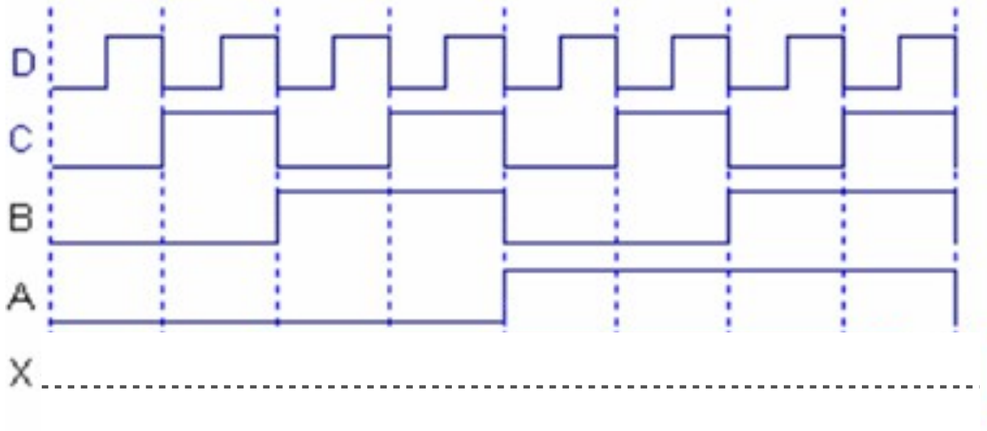
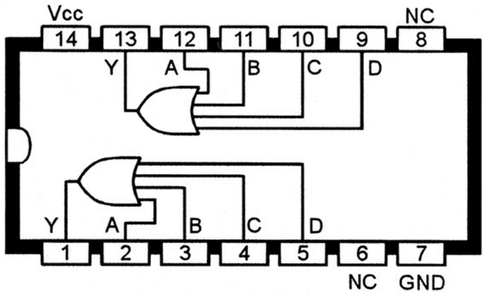


Diagrama de formas de onda

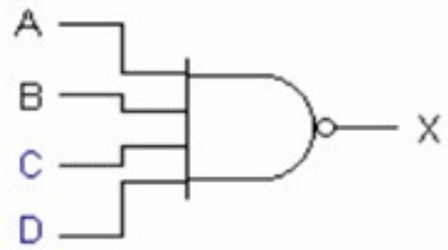
A	B	C	D	X
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	



Exemplo:

NAND(4)

- Que porta é esta?



$X = \overline{ABCD}$

Note:

$x \cdot 0 = 0$

$x \cdot 1 = x$

A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

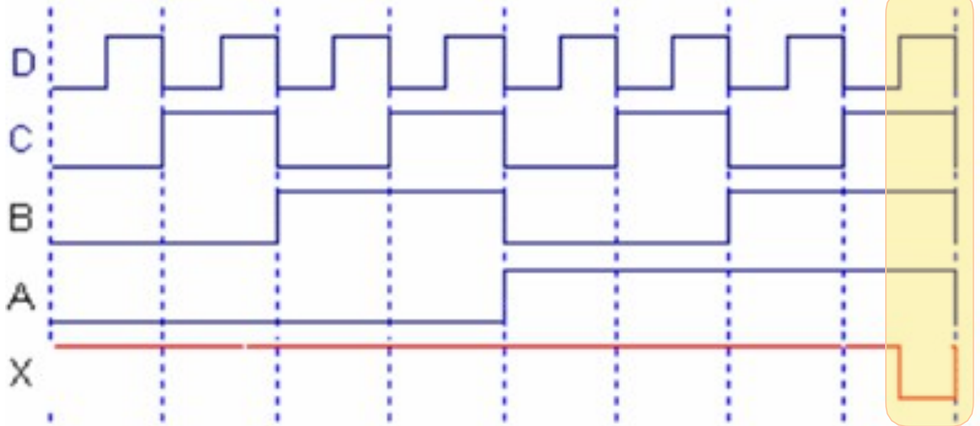
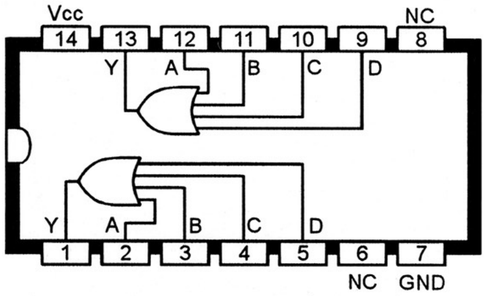


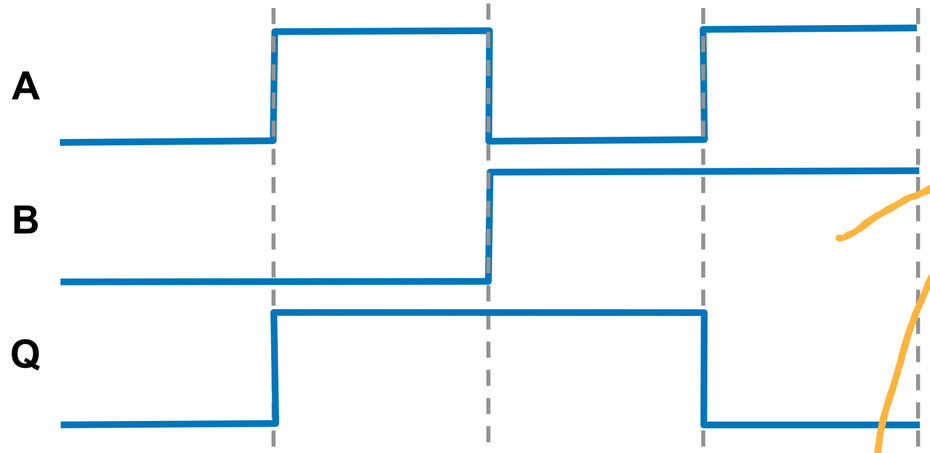
Diagrama de formas de onda



Últimas portas: XOR e NXOR

- XOR: eXclusive OR:

XOR				INPUT		OUTPUT
				A	B	Q
				0	0	0
				0	1	1
				1	0	1
				1	1	0

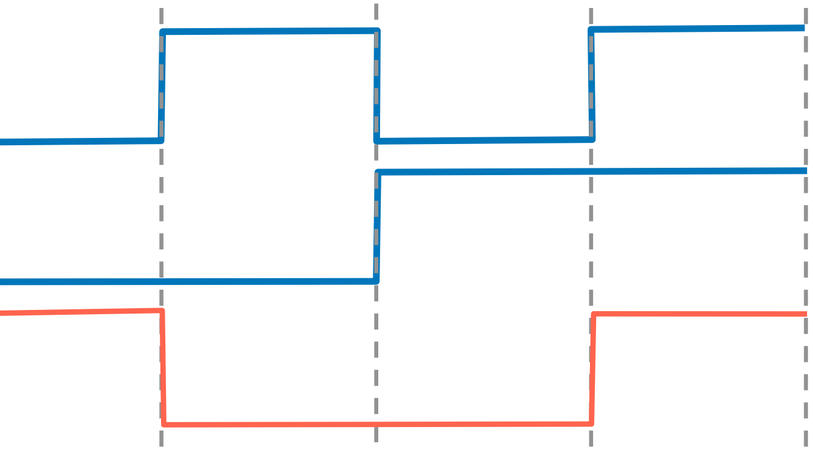


$A \oplus B$ or $A \veebar B$

Detector de Desigualdades

- NXOR: Negative eXclusive OR:

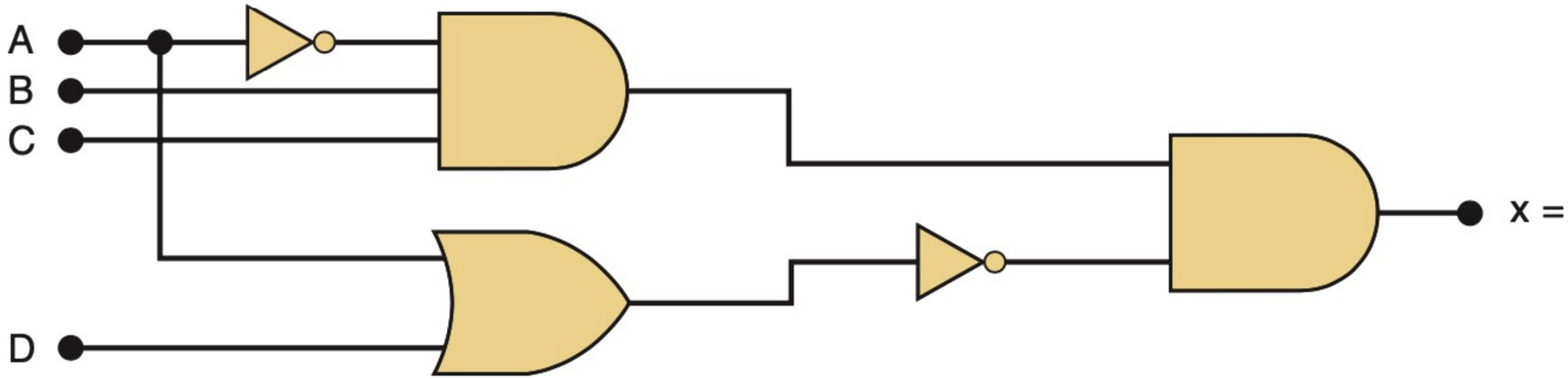
XNOR				INPUT		OUTPUT
				A	B	Q
				0	0	1
				0	1	0
				1	0	0
				1	1	1



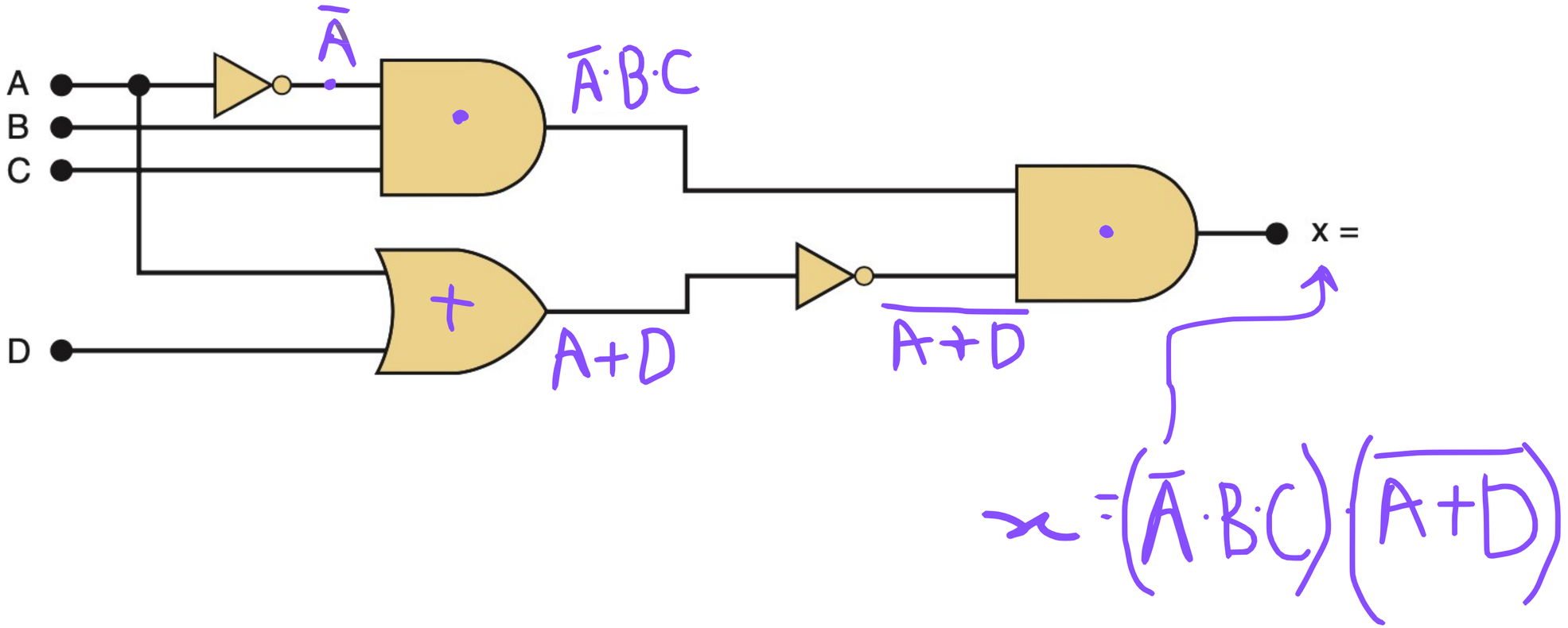
$\overline{A \oplus B}$ or $A \odot B$

Detector de Igualdades

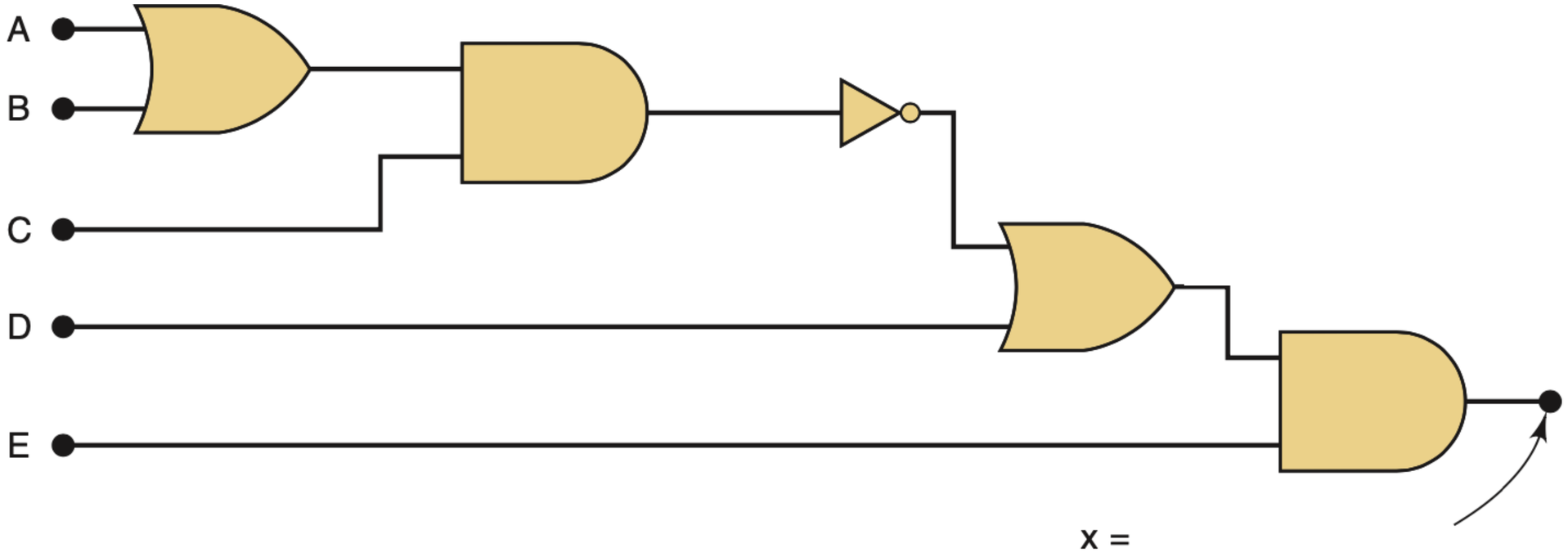
Deduzindo equações:



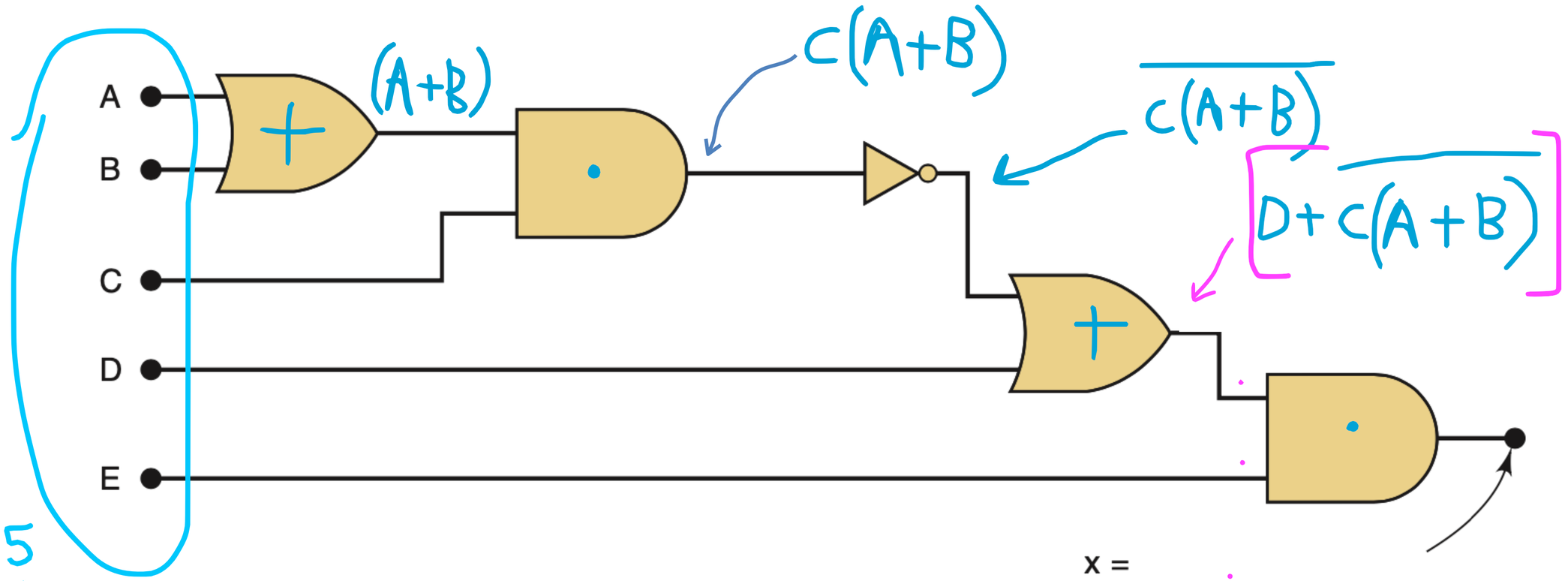
Deduzindo equações:



Deduzindo equações:



Deduzindo equações:



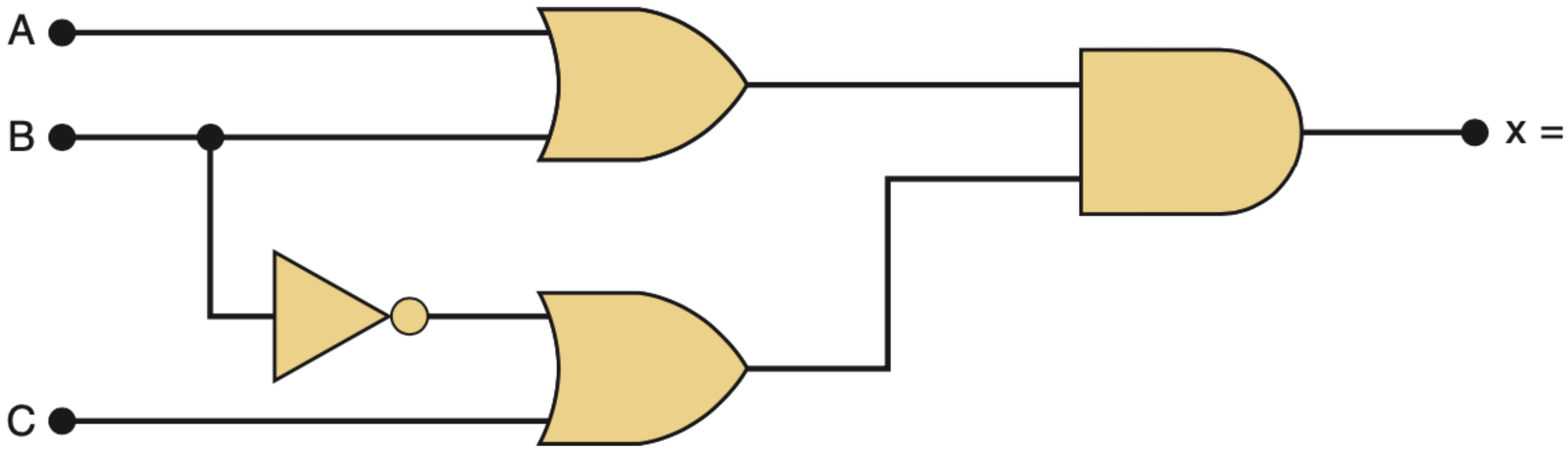
5

$2^5 = 32$

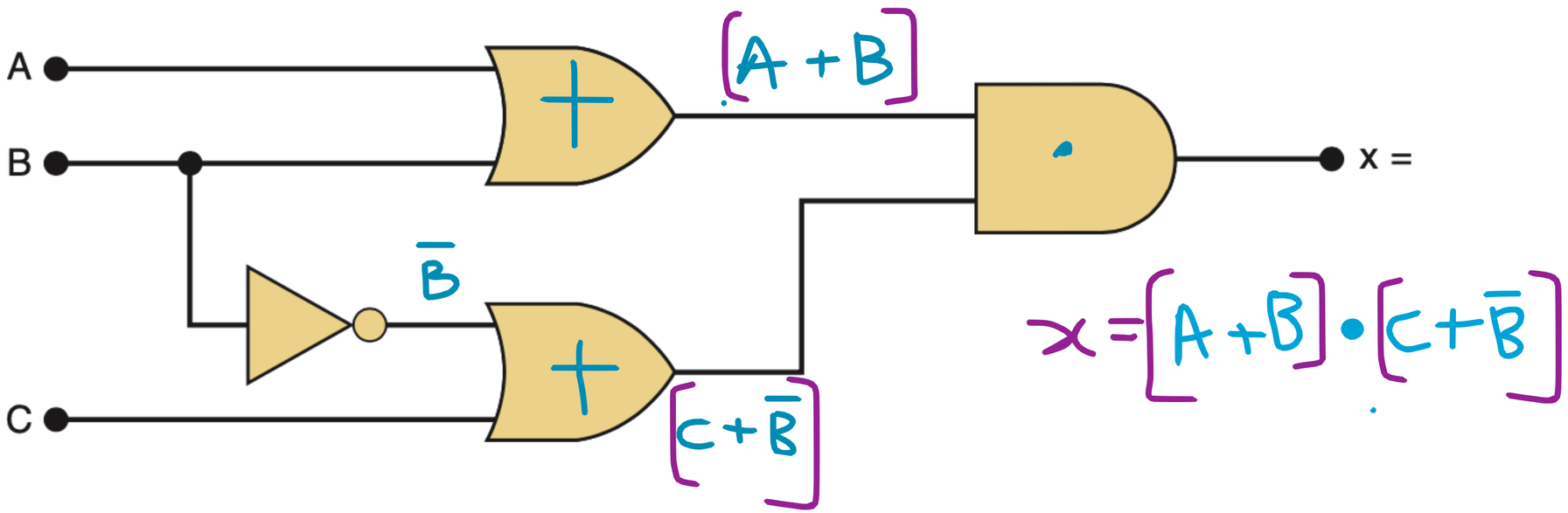
$= 1?$

$x = E \cdot [D + \overline{C(A+B)}]$

Deduzindo equações:



Deduzindo equações:



Desenhando circuitos:

- a) $x = \overline{A}BC\overline{(A + D)}$
- b) $y = AC + B\overline{C} + \overline{\overline{A}BC}$
- c) $z = [D + \overline{(A + B)}C]E$

Desenhando circuitos:

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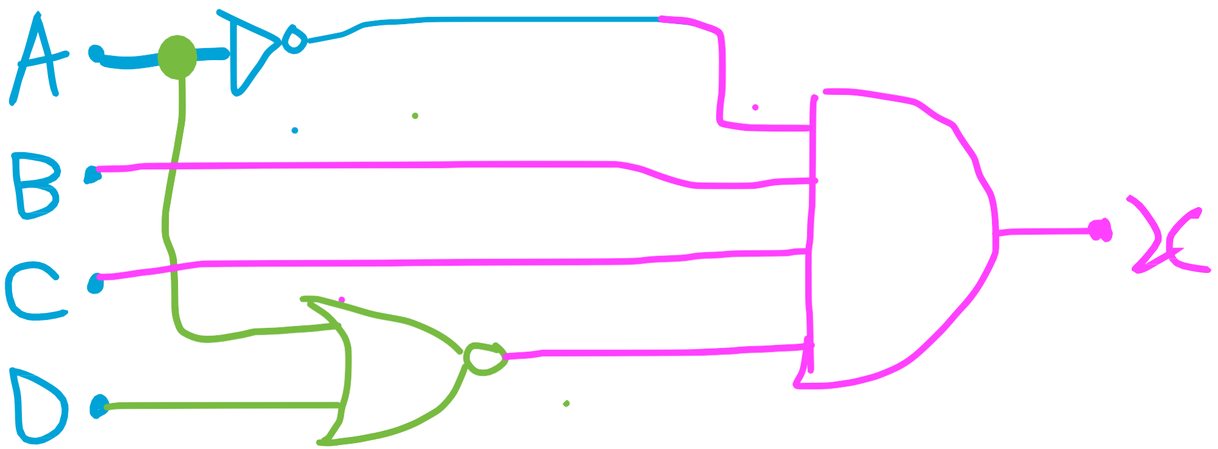
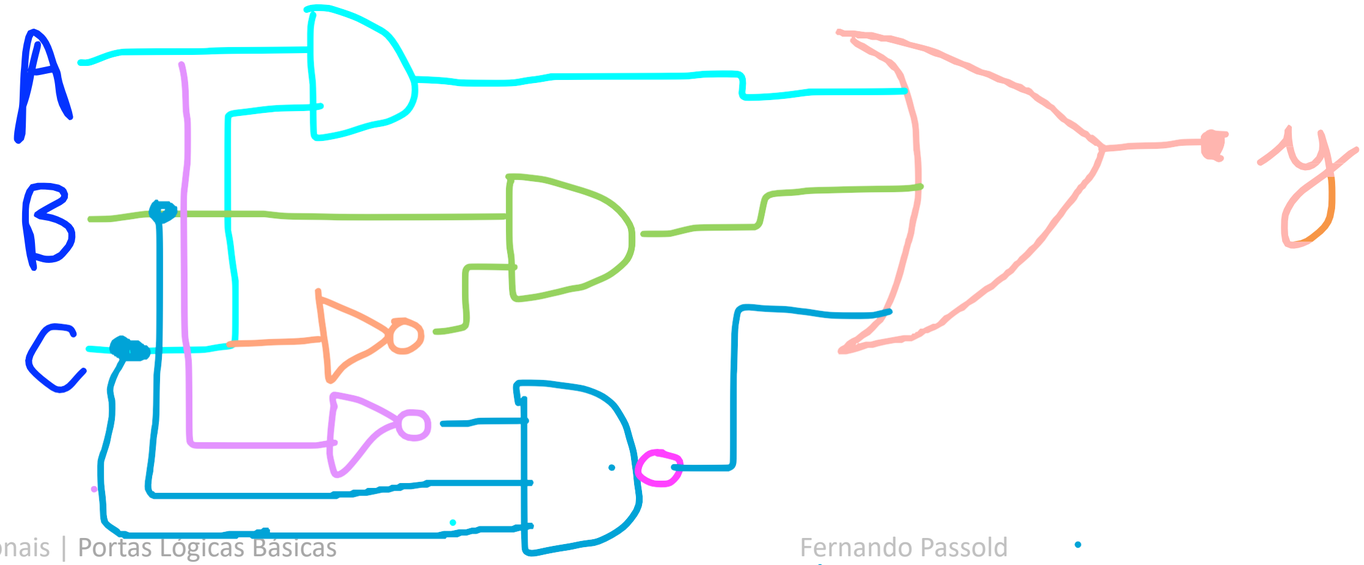


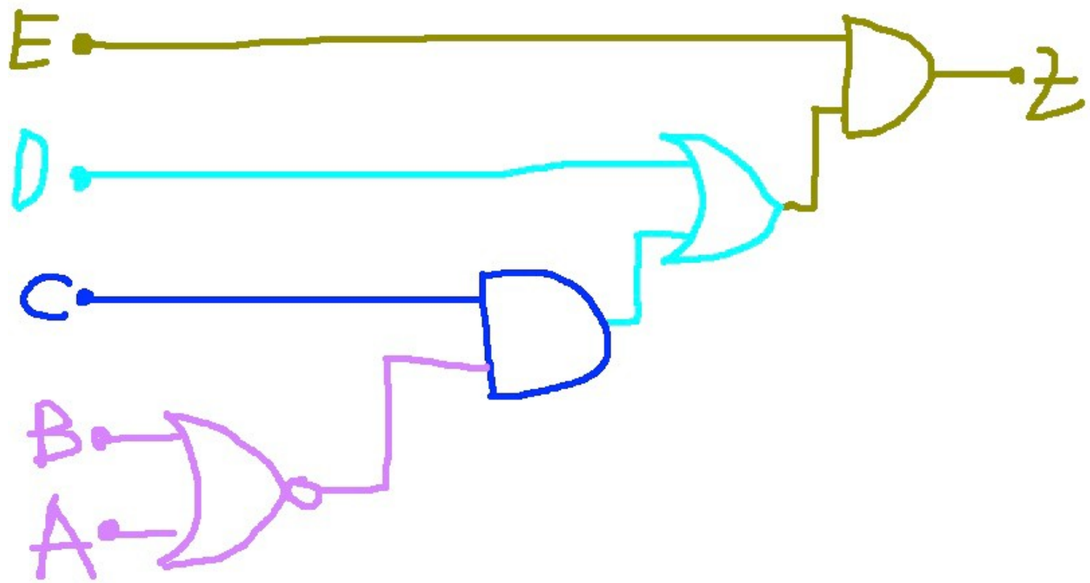
Desenhando circuitos:

- a) $x = \overline{A}\,BC\,\overline{(A + D)}$
- b) $y = AC + B\,\overline{C} + \overline{\overline{A}\,BC}$
- c) $z = \overline{[D + (A + B)\,C]\,E}$



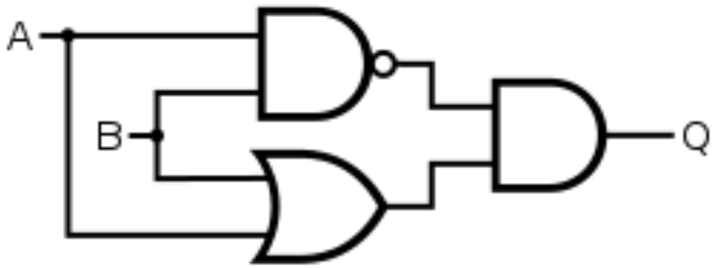
Desenhando circuitos:

- a) $x = \overline{A}\,BC\,\overline{(A + D)}$
- b) $y = \overline{A}C + B\,\overline{C} + \overline{\overline{A}\,BC}$
- c) $z = [D + (A + B)\,C]\,E$



Deduzindo expressões e tabelas verdade

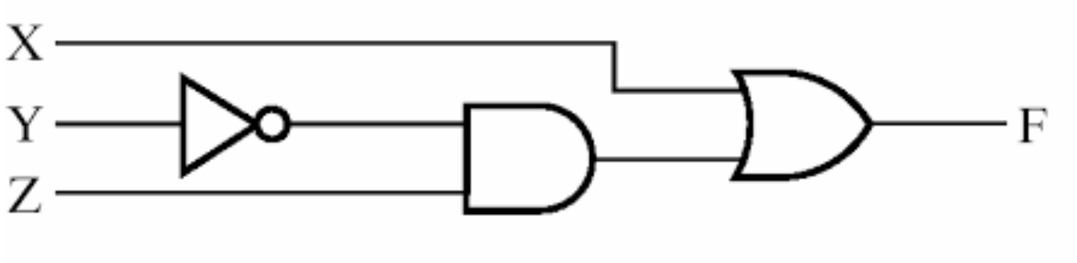
- Quando Q comuta para nível lógico ALTO?



$$Q = (A + B) \cdot (\bar{A} + \bar{B})$$

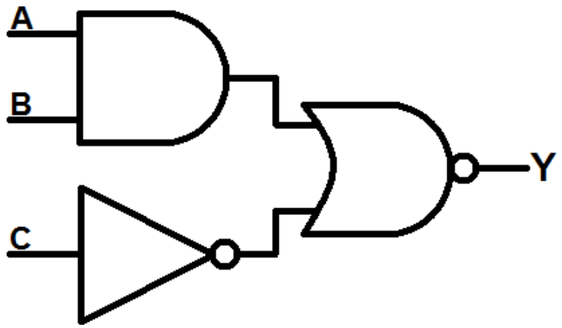
Deduzindo expressões e tabelas verdade

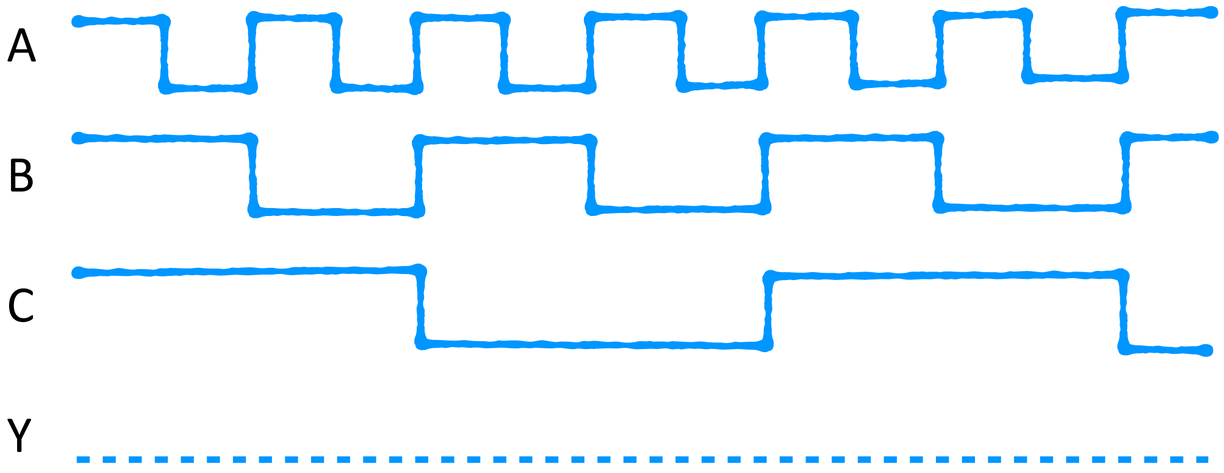
- Quando F comuta para nível lógico ALTO?



Deduzindo expressões e tabelas verdade

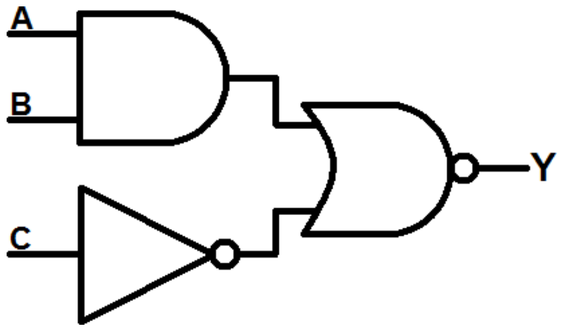
- Quando Y comuta para nível lógico ALTO?

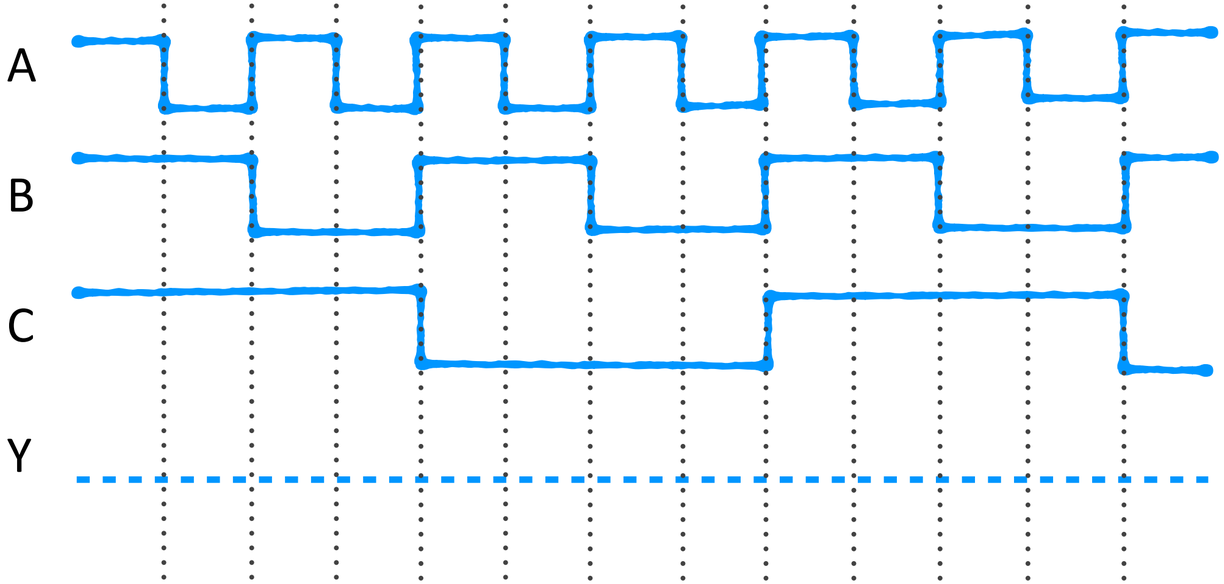




Deduzindo expressões e tabelas verdade

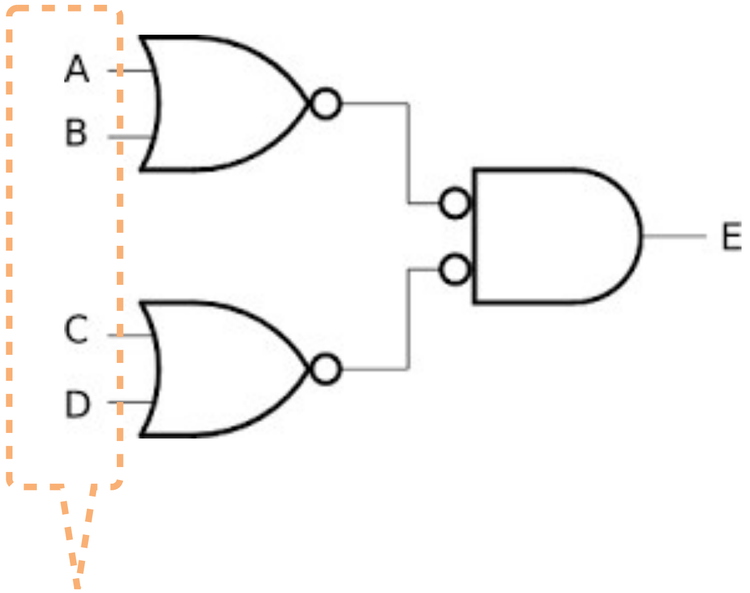
- Quando Y comuta para nível lógico ALTO?

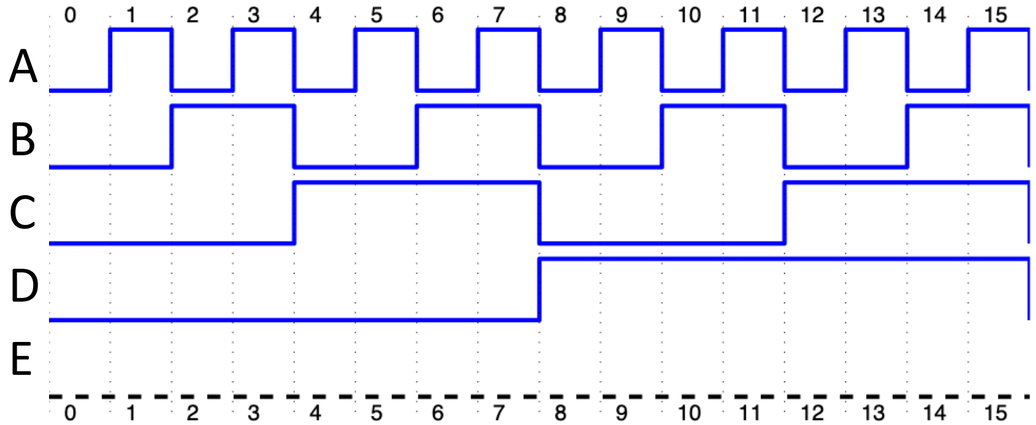




Deduzindo expressões e tabelas verdade

- Complete a forma de onda na saída E (deduza a tabela verdade para E):

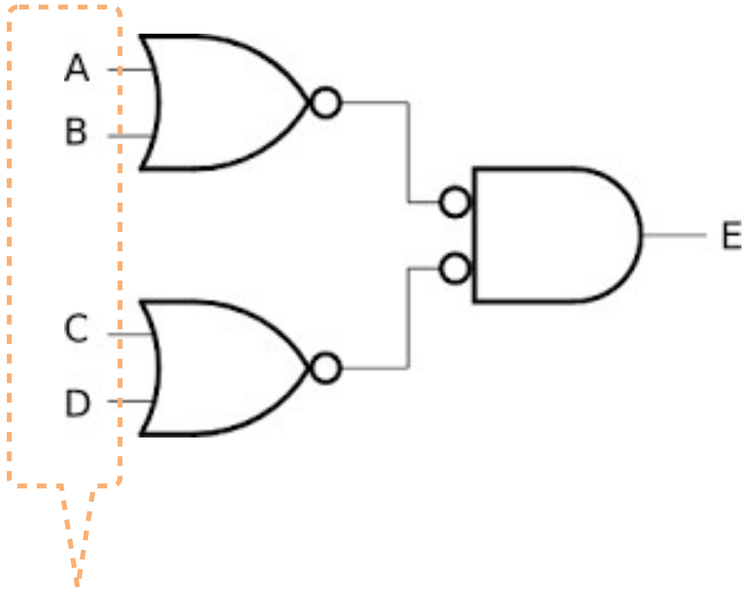


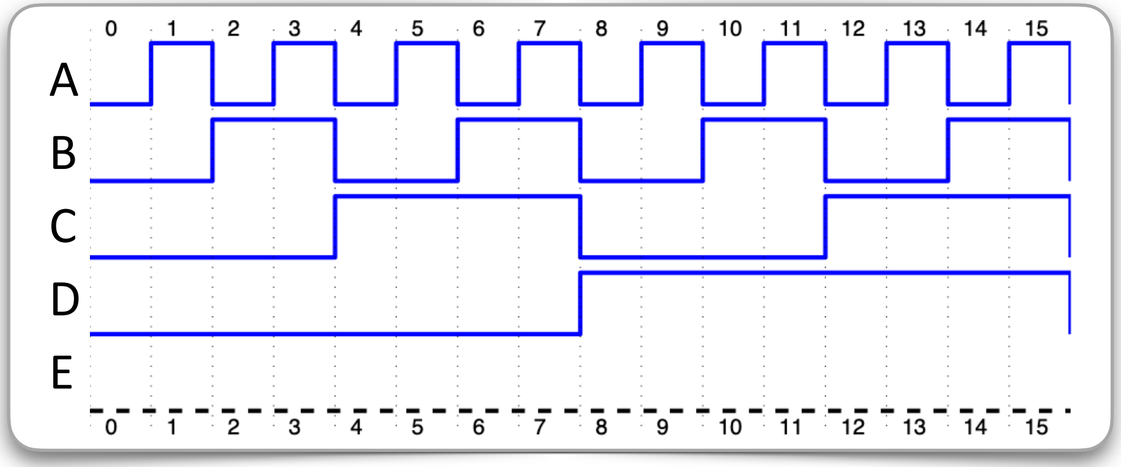


4 variáveis de entrada $\Rightarrow 2^4 = 16$ combinações de entrada

Deduzindo expressões e tabelas verdade

- Complete a forma de onda na saída E (deduza a tabela verdade





4 variáveis de entrada $\Rightarrow 2^4 = 16$ combinações de entrada

Deduzindo expressões e tabelas verdade

- Quando Q comuta para nível lógico ALTO?